

# PRIME NUMBERS, EXPLICIT FORMULAS AND THE RIEMANN HYPOTHESIS

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*“To see form where form is yet unmade  
And shape in the ephemeral air;  
... To see light where others had gone to sleep  
And a path through the trackless woods;  
... In whose presence, things dare not be incoherent,  
This inspired mind is the visionary eye.” \*\**

Euler first noticed the connection between prime numbers and analysis. He showed that the unique factorization of natural numbers is equivalent to the analytic statement

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1} \quad (s > 1)$$

where the infinite product is over the prime numbers. Euler noted that since the harmonic series diverges, the identity implies the infinitude of primes. This simple, yet profound, observation was utilised in 1837, by Dirichlet, to resolve a question of antiquity: the infinitude of primes in a primitive arithmetic progression. We can say that analytic number theory was created in the 1837 memoir of Dirichlet.

In 1860, Riemann showed that the key to a deeper investigation of the prime numbers lies in the study of

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

as a function of a complex variable  $s$ . Originally defined for  $\Re(s) > 1$ , Riemann showed that  $\zeta(s)$  can be analytically continued for all complex values of  $s$  except for a simple pole at  $s = 1$ . Moreover, he proved that  $\zeta(s)$  satisfies the functional equation:

$$\pi^{-s/2} \Gamma(s/2) \zeta(s) = \pi^{-\frac{(1-s)}{2}} \Gamma\left(\frac{1-s}{2}\right) \zeta(1-s).$$

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\*\* From the unpublished writings of VKM.

Riemann then went on to make a number of seminal conjectures:

- (1)  $\zeta(s)$  has infinitely many zeroes in  $0 \leq \Re(s) \leq 1$ .
- (2) The number of zeroes of  $\zeta(s)$  satisfying  $0 \leq \Re(s) \leq 1$ ,  $0 \leq \Im(s) \leq T$  is

$$\frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T),$$

as  $T \rightarrow \infty$ .

- (3) The function

$$\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

has a product representation

$$e^{A+Bs} \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho}$$

where  $A$  and  $B$  are constants and  $\rho$  runs through the zeroes of  $\zeta(s)$  in the *critical strip*  $0 \leq \Re(s) \leq 1$ .

- (4) Let us define  $\Lambda(n) = \log p$  whenever  $n$  is a power of a single prime  $p$  and zero otherwise. Set

$$\psi(x) = \sum_{n \leq x} \Lambda(n).$$

Then,

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \frac{\zeta'}{\zeta}(0) - \frac{1}{2} \log(1 - x^{-2}).$$

- (5) (The famous Riemann hypothesis) All zeros  $\rho$  of  $\zeta(s)$  in the critical strip satisfy  $\Re(\rho) = \frac{1}{2}$ .

The next four decades after 1860 were devoted to the proofs of these conjectures.

(1), (2), and (4) were proved by von Mangoldt in 1895 and (3) was proved in 1893 by Hadamard as a consequence of his theory of infinite products for analytic functions. Only (5) has defied all attempts at a proof. The most compelling reason for believing in the truth of (5) comes from the analogues in algebraic geometry formulated by E. Artin and A. Weil and subsequently proved by H. Hasse, A. Weil, A. Grothendieck and P. Deligne in a succession of remarkable conceptual milestones of mathematics.

(4) is interesting in that it gives an explicit relationship between the zeroes of  $\zeta(s)$  and the prime numbers. Indeed, in 1893, Hadamard and de la vallée Poussin showed that  $\zeta(s) \neq 0$  for  $\Re(s) = 1$  and thereby deduced from (4) that

$$\psi(x) \sim x$$

as  $x \rightarrow \infty$ . From this, it is straightforward to infer that the number of primes up to  $x$  satisfies the asymptotic law

$$\sim \frac{x}{\log x}$$

as  $x \rightarrow \infty$ , a fact first conjectured by Gauss at the age of seventeen.

This lecture is about (4) and its generalisations by Weil. In 1953, Weil outlined a method by which the Riemann hypothesis can be proved. He established an explicit formula relating a suitable function  $f$  evaluated at the logarithms of the prime numbers to the Fourier transform of  $f$  evaluated at the zeroes of  $\zeta(s)$  and showed that the positivity of the corresponding sum for a certain class of functions  $f$  is equivalent to the Riemann hypothesis. He had successfully utilised such an approach in establishing the geometric analogue alluded to above. I will go into this in more detail below. But first, I would like to describe some early attempts in trying to prove the Riemann hypothesis.

In 1919, Harald Bohr developed a method for studying the set of values taken by a Dirichlet series in a half plane. His first step consisted of introducing an equivalence relation among the Dirichlet series. He then showed that equivalent Dirichlet series take the same set of values in certain half planes. Although his theory is more general, I will illustrate it only in a special case.

I begin by stating the following theorem of Turán.

**THEOREM 1.** (P. Turán) If each of the finite series

$$\zeta_N(s) = \sum_{n \leq N} \frac{1}{n^s}$$

does not vanish in  $\Re(s) > 1$  then the Riemann hypothesis is true.

The proof of this fact uses Bohr's theory and a classical theorem of Landau. We say that the Dirichlet series

$$f(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

has *abscissa of absolute convergence*  $\sigma_a$  if for  $\Re(s) > \sigma_a$  the series converges absolutely. We can also consider integrals of the form

$$F(s) = \int_1^{\infty} \frac{A(x)}{x^s} dx$$

and define the *abscissa of convergence* of  $F(s)$  as the infimum of all the  $\sigma$  such that integral converges for  $\Re(s) > \sigma$ . In addition to Bohr's theory, a key ingredient in the proof of Theorem 1 is a classical theorem of Landau, which we state in the form we need it.

**THEOREM 0.** (E. Landau) Let  $F(s)$  be a meromorphic function. If in some half plane,

$$F(s) = \int_1^\infty \frac{A(x)}{x^s} dx$$

and  $A(x) \geq 0$  for all  $x \geq 1$ , then the largest real singularity of  $F(s)$  is the abscissa of convergence of the integral.

We now define an equivalence relation amongst Dirichlet series. Two Dirichlet series

$$f(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s} \quad g(s) = \sum_{n=1}^{\infty} \frac{b_n}{n^s}$$

are said to be *equivalent* if and only if there is a completely multiplicative function  $h$  (that is,  $h(mn) = h(m)h(n)$ ) such that

- (a)  $a_n = h(n)b_n$  for all  $n \geq 1$ , and
- (b) if  $a_n \neq 0$ ,  $|h(p)| = 1$  for every prime divisor  $p$  of  $n$ .

It is clear that equivalent Dirichlet series have the same abscissa of absolute convergence. Bohr proved the following theorem.

**THEOREM 2.** (H. Bohr) If  $f$  and  $g$  are equivalent Dirichlet series with abscissa of absolute convergence  $\sigma_a$ , then in any half plane  $\Re(s) > \sigma_1$ , with  $\sigma_1 \geq \sigma_a$ , the functions  $f(s)$  and  $g(s)$  take on the same set of values.

We shall use Theorem 2 to prove Theorem 1. To this end, let us define the Liouville function  $\lambda$  as

$$\lambda(n) = (-1)^\alpha$$

where  $\alpha$  is the total number of prime factors of  $n$  counting multiplicity. It is obvious that  $\lambda$  is a completely multiplicative function. In view of this, it satisfies the identity

$$(1) \quad \frac{\zeta(2s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\lambda(n)}{n^s} \quad (\Re(s) > 1).$$

This identity can be written in integral form using partial summation. Indeed, let

$$S(x) = \sum_{n \leq x} \frac{\lambda(n)}{n}.$$

Then,

$$\sum_{n=1}^{\infty} \frac{\lambda(n)}{n^{s+1}} = s \int_1^{\infty} \frac{S(t)}{t^{s+1}} dt \quad (\Re(s) > 0).$$

If we change  $s$  into  $s - 1$ , we get

$$\sum_{n=1}^{\infty} \frac{\lambda(n)}{n^s} = (s - 1) \int_1^{\infty} \frac{S(t)}{t^s} dt \quad (\Re(s) > 1).$$

In view of (1), we obtain

$$\frac{\zeta(2s)}{(s - 1)\zeta(s)} = \int_1^{\infty} \frac{S(t)}{t^s} dt.$$

Since  $\lambda$  is completely multiplicative, the (finite) Dirichlet series

$$\sum_{n \leq N} \frac{1}{n^s} \quad \sum_{n \leq N} \frac{\lambda(n)}{n^s}$$

are equivalent. Thus, if  $\zeta_N(s) \neq 0$  in  $\Re(s) > 1$ , then by Bohr's theorem (Theorem 2) we find

$$\sum_{n \leq N} \frac{\lambda(n)}{n^s} \neq 0 \quad (\Re(s) > 1).$$

Letting  $s \rightarrow 1^+$  we find

$$\sum_{n \leq N} \frac{\lambda(n)}{n} \geq 0$$

in view of the continuity of the function under consideration. Therefore,  $S(t) \geq 0$  for all values of  $t \geq 1$ . By Landau's theorem (Theorem 0), the abscissa of convergence of

$$(2) \quad \int_1^{\infty} \frac{S(t)}{t^s} dt$$

is the largest *real* singularity of the meromorphic function

$$\frac{\zeta(2s)}{(s - 1)\zeta(s)}.$$

The only *real* singularity of this function is at  $s = 1/2$  arising from the simple pole of  $\zeta(2s)$ . Hence, (2) converges for  $\Re(s) > 1/2$ . Hence,

$$\frac{\zeta(2s)}{(s - 1)\zeta(s)}$$

is analytic for  $\Re(s) > 1/2$ , which implies that  $\zeta(s) \neq 0$  for  $\Re(s) > 1/2$ . This is the Riemann hypothesis.

Unfortunately, Turán's hypothesis is false. It was shown by H. Montgomery that  $\zeta_N(s)$  has a zero in the half-plane  $\Re(s) > 1$  for every  $N$  sufficiently large. A closer examination of the proof reveals that we only utilised the positivity of  $S(t)$ . However, Haselgrove, in 1958 showed that  $S(t)$  is negative for infinitely many values of  $t$ .

The theme of positivity also enters into Weil's equivalent formulation of the Riemann hypothesis.

Let

$$\rho = \frac{1}{2} + i\gamma, \quad \gamma \in \mathbb{C}$$

be a typical non-trivial zero of  $\zeta(s)$ , and let  $h(r)$  be any analytic function on  $|Im(r)| \leq \frac{1}{2} + \delta$  such that

$$(*) \quad h(-r) = h(r), \quad \text{and} \quad |h(r)| \leq A(1 + |r|)^{-2-\delta} \quad (A > 0, \delta > 0).$$

Let

$$g(u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} h(r) e^{-iru} dr$$

be its Fourier transform. Then, we have:

### Weil's explicit formula:

Let  $h$  satisfy  $(*)$ . Then,

$$(1) \quad \sum_{\gamma} h(\gamma) = \frac{1}{2\pi} \int_{-\infty}^{\infty} h(r) \frac{\Gamma'}{\Gamma} \left( \frac{1}{4} + \frac{1}{2}ir \right) dr + h(i/2) + h(-i/2) - g(0) \log \pi - 2 \sum_{n=1}^{\infty} \frac{\Lambda(n)}{\sqrt{n}} g(\log n).$$

The sum is over all non-trivial zeroes of  $\zeta(s)$ . This formula has an important connection with the Riemann hypothesis. Suppose that

$$g(u) = \int_{-\infty}^{\infty} g_0(t+u) \overline{g_0(t)} dt \quad (g_0 \text{ even}).$$

Then, observe that  $h(r) = h_0(r) \overline{h_0(\bar{r})}$  for  $|Im(r)| \leq \frac{1}{2} + \delta$ . Weil showed that the Riemann hypothesis is equivalent to showing that (1) is positive for all such  $g$ .

In 1950, Selberg noticed a remarkable similarity between this formula and a similar one, which he derived connecting the eigenvalues of the Laplacian and the lengths of closed geodesics on Riemannian manifolds.

Indeed, let  $F$  be a compact Riemann surface of genus  $g \geq 2$ . The universal covering space of  $F$  is conformally equivalent to the upper half plane

$$H = \{x + iy : y > 0\}.$$

We can express  $F$  in the form  $\Gamma \backslash H$  where  $\Gamma$  is a strictly hyperbolic Fuchsian group. The group  $\Gamma \subseteq PSL_2(\mathbf{R})$  will have a compact fundamental polygon  $\mathcal{F}$ . Recall that  $\Gamma \simeq \pi_1(F)$ , the fundamental group of  $F$ . There is the hyperbolic (Poincaré) metric on  $H$ :

$$ds = \frac{|dz|}{y},$$

whose corresponding area element is

$$d\mu(z) = \frac{dx dy}{y^2}, \quad z = x + iy.$$

The hyperbolic distance is given by

$$d(z_1, z_2) = \arg \cosh \left[ 1 + \frac{|z_1 - z_2|^2}{2y_1 y_2} \right].$$

It is easily checked that this distance is invariant under the action of  $SL_2(\mathbf{R})$ . We consider

$$L^2(\Gamma \backslash H) = \{f : H \rightarrow \mathbb{C} : \int_{\mathcal{F}} |f(z)|^2 d\mu(z) < \infty, \quad f(\gamma z) = f(z) \quad \forall \gamma \in \Gamma, z \in H\}.$$

Such functions are called *automorphic*. We can introduce an inner product on  $L^2(\Gamma \backslash H)$  as follows:

$$(f_1, f_2) = \int_{\mathcal{F}} f_1(z) \overline{f_2(z)} d\mu(z).$$

The Laplace operator is the non-Euclidean Laplacian:

$$\Delta = y^2 \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right).$$

We write  $\lambda_0 \leq \lambda_1 < \dots$  for the sequence of eigenvalues and  $\phi_0, \phi_1, \dots$  for the corresponding eigenfunctions.

Let  $\sigma$  be an element of  $PSL_2(\mathbf{R})$ , which is not the identity. Then, the Jordan canonical form of  $\sigma$  is one of the following:

$$\begin{aligned} (i) & \begin{pmatrix} \omega & \\ & \bar{\omega} \end{pmatrix}, \quad \omega \in \mathbb{C}, \quad |\omega| = 1 \\ (ii) & \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix}, \quad t \in \mathbf{R} \\ (iii) & \begin{pmatrix} 1 & x \\ & 1 \end{pmatrix}, \quad x \in \mathbf{R} \end{aligned}$$

$\sigma$  is called *elliptic*, *hyperbolic*, *parabolic* according as the Jordan canonical form of  $\sigma$  is of type (i), (ii), (iii) respectively.

Now let us consider the hyperbolic conjugacy classes of  $\Gamma$ . We define the *norm* of  $P$  as  $t^2$  and write  $N(P) = t^2$ . Moreover, the centraliser of  $P$  in  $\Gamma$ , denoted  $C_\Gamma(P)$ , must reduce to a cyclic group since  $\Gamma$  acts discontinuously on the upper half plane. Indeed, it is easily verified that  $C_\Gamma(P)$  is generated by a primitive hyperbolic element  $P_0$ . Thus, if  $P$  is a hyperbolic element,  $P = P_0^k$  for some integer  $k$ , and  $P_0$  is a unique primitive hyperbolic element. We define  $\Lambda(P) = \log N(P_0)$ . Selberg proved the following:

**Trace formula:**

Let  $h(r)$  denote any analytic function on  $|Im(r)| \leq \frac{1}{2} + \delta$  such that

$$h(-r) = h(r) \quad \text{and} \quad |h(r)| \leq A(1 + |r|)^{-2-\delta}, \quad (A > 0, \delta > 0).$$

Then,

$$\sum_{n=0}^{\infty} h(r_n) = \frac{vol(\mathcal{F})}{4\pi} \int_{-\infty}^{\infty} rh(r) \tanh \pi r dr + \sum_P \frac{\Lambda(P)}{N(P)^{1/2} - N(P)^{-1/2}} g(\log N(P))$$

where the sum over  $P$  is taken over all hyperbolic conjugacy classes and  $s_n = \frac{1}{2} + ir_n$ ,  $\lambda_n = s_n(1 - s_n)$ .

Selberg found a zeta function which he denoted by  $Z(s)$ , and which corresponds to this formula in the same way as Weil's formula corresponds to the Riemann zeta function. By means of his trace formula, he showed that the non-trivial zeroes of  $Z(s)$  are located on the line  $Re s = \frac{1}{2}$ . Perhaps this analogy would give a better understanding of the elusive Riemann hypothesis.

The symmetry of the functional equation may have led Riemann to make his famous hypothesis. Yet, this is unlikely. It is easy to produce Dirichlet series with functional equations of a similar type but which violate the analogue of the Riemann hypothesis. Many of these examples arise from the theory of Eisenstein series or Dirichlet series which are Mellin transforms of modular forms. These counterexamples, however, do not possess an Euler product and therefore, this fact must enter into any proof of the Riemann hypothesis.

The proof of Weil's explicit formula is not hard. Let  $h(r)$  be an analytic function satisfying the properties

$$(*) \quad h(r) = h(-r), \quad |h(r)| \leq A(1 + |r|)^{-2-\delta}.$$

We will prove:

**PROPOSITION:** (Weil)

$$\sum_{\gamma} h(\gamma) = \frac{1}{2\pi} \int_{-\infty}^{\infty} h(r) \frac{\Gamma'}{\Gamma} \left( \frac{1}{4} + \frac{1}{2}ir \right) dr + 2h(i/2) - g(0) \log \pi - 2 \sum_{n=1}^{\infty} \frac{\Lambda(n)}{\sqrt{n}} g(\log n)$$

where the first sum is over non-trivial zeroes  $\frac{1}{2} + i\gamma$ ,  $\gamma \in \mathbb{C}$ , of  $\zeta(s)$ . Moreover,

$$g(u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} h(r) e^{-iru} dr.$$

**Proof.** To prove this formula, it is convenient to define

$$H(s) = \int_{-\infty}^{\infty} g(u) e^{(s-\frac{1}{2})u} du.$$

Then,

$$H\left(\frac{1}{2} + iw\right) = h(w)$$

by the Fourier inversion formula. Since  $h$  is even,  $H(w) = H(1-w)$ . As  $H$  is essentially the Laplace transform of  $g$ , it is useful to recall the Mellin inversion formula:

$$g(u) = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} H(w) e^{(w-\frac{1}{2})u} dw.$$

In order to prove the explicit formula, consider the integral

$$\frac{1}{2\pi i} \int_{\partial R} H(s) \frac{\zeta'}{\zeta}(s) ds$$

where

$$R = [-\delta, 1 + \delta] \times [-T, T]$$

where  $T$  is suitably chosen as an element of a sequence of  $T_m$  such that  $T_m \rightarrow \infty$  and  $\zeta(s) \neq 0$  for  $\text{Im}(s) = T_m$ . The singularities of the integrand are the non-trivial zeroes of  $\zeta(s)$ . By Cauchy's theorem, the integral is

$$\sum_{|\text{Im}(i\gamma)| < T} h(\gamma).$$

It can also be arranged that for these  $T_m$ ,

$$\frac{\zeta'}{\zeta}(s) = O(\log^2(|t| + 1))$$

on the horizontal and vertical lines of the contour. The growth condition on  $h$  is sufficient to ensure that the contribution from the horizontal integrals tends to 0 as  $T_m \rightarrow \infty$ . Now by the functional equation,

$$-\frac{\zeta}{\zeta}(1-s) = -\log \pi + \frac{1}{2} \frac{\Gamma'}{\Gamma}(s/2) + \frac{1}{2} \frac{\Gamma'}{\Gamma}((1-s)/2) + \frac{\zeta'}{\zeta}(s)$$

so that one of the vertical integrals is transformed into the other vertical integral:

$$-\frac{1}{2\pi i} \int_{-\delta+i\infty}^{-\delta-i\infty} H(s) \frac{\zeta'}{\zeta}(s) ds = - \int_{1+\delta-i\infty}^{1+\delta+i\infty} H(1-w) \frac{\zeta'}{\zeta}(1-w) dw$$

which is

$$= \int_{1+\delta-i\infty}^{1+\delta+i\infty} H(w) \left\{ -\log \pi + \frac{1}{2} \frac{\Gamma'}{\Gamma}(w/2) + \frac{1}{2} \frac{\Gamma'}{\Gamma}((1-w)/2) + \frac{\zeta'}{\zeta}(w) \right\} dw.$$

The part of the integral involving the zeta function gives for each vertical integral:

$$- \sum_{n=1}^{\infty} \frac{\Lambda(n)}{\sqrt{n}} \int_{1+\delta-i\infty}^{1+\delta+i\infty} H(w) e^{(w-\frac{1}{2}) \log n} dw = - \sum_{n=1}^{\infty} \frac{\Lambda(n)}{\sqrt{n}} g(\log n),$$

by the Mellin inversion formula. We also obtain the constant term

$$-g(0) \log \pi.$$

To evaluate the integrals involving the gamma function, we move the line of integration to  $\operatorname{Re}(w) = \frac{1}{2}$ . This is valid since

$$\frac{\Gamma'}{\Gamma}(w) = O(\log(|w| + 2))$$

in the half plane under consideration. This completes the proof of the proposition.

The most interesting consequence of the explicit formula is the following equivalent formulation of the Riemann hypothesis.

**COROLLARY.** Suppose that

$$g(u) = \int_{-\infty}^{\infty} g_0(t+u) \overline{g_0(t)} dt$$

for some  $g_0$ . Then the Riemann hypothesis is true if and only if

$$\sum_{\gamma} \hat{g}(\gamma) \geq 0$$

for all such  $g$ . (Here, the sum is over all the non-trivial zeroes  $\frac{1}{2} + i\gamma$ ,  $\gamma \in \mathbb{C}$  of  $\zeta(s)$ .)

**Proof.** We can write

$$g(u) = g_0(u) * \overline{g_0(-u)}.$$

If we set

$$g_1(u) = \overline{g_0(-u)},$$

then, a simple calculation shows that

$$\begin{aligned} \hat{g}_1(r) &= \int_{-\infty}^{\infty} \overline{g_0(-u)} e^{-iru} du \\ &= \overline{\int_{-\infty}^{\infty} g_0(-u) e^{-i\bar{r}u} du} \\ &= \overline{h_0(\bar{r})} \end{aligned}$$

where  $h_0 = \hat{g}_0$ . Thus, by the theory of the Fourier transform,

$$h(r) = h_0(r) \overline{h_0(\bar{r})}.$$

If all the zeroes  $\rho$  of  $\zeta(s)$  have  $Re(\rho) = \frac{1}{2}$ , then  $\gamma$  is real and

$$h(\gamma) = |h_0(\gamma)|^2 \geq 0$$

and hence

$$\sum_{\gamma} h(\gamma) \geq 0.$$

Conversely, suppose that the sum is non-negative for all such functions and there is a zero  $\rho_0 = \beta_0 + i\gamma_0$  with  $\beta_0 \neq \frac{1}{2}$ . We will utilise the fact that the Fourier transform of a function of the form

$$P(x)e^{-Ax^2},$$

where  $P$  is a polynomial and  $A > 0$ , is again a function of the same form. We will set

$$h_0(x) = P(x)e^{-Ax^2}$$

for a suitable  $P$  and  $A$ , and show that for such a choice,

$$\sum_{\gamma} h(\gamma) < 0.$$

This will establish the corollary. Indeed, set

$$z = i(s - \frac{1}{2} - i\gamma_0)$$

and let  $\eta = \eta_\rho = i(\rho - \frac{1}{2} - i\gamma_0)$ . Then,  $\eta_0 = i(\beta_0 - \frac{1}{2})$  corresponds to  $\rho = \rho_0$ . Let  $Q(x)$  be the polynomial having for zeroes all the  $\eta$ 's satisfying  $|Re(\eta)| \leq 2$  except for  $\eta_0, \bar{\eta}_0$ . (Note that if  $\rho$  is a zero of  $\zeta(s)$ , then  $\bar{\rho}$  and  $1 - \rho$ ,  $1 - \bar{\rho}$  are zeroes of  $\zeta(s)$  by virtue of the functional equation and the fact that  $\zeta(s)$  is real for real values of  $s$ .) Thus, the zeroes of  $Q(x)$  are either real or occur in conjugate pairs. Hence,  $Q$  has real coefficients. We will set

$$P(x) = xQ(x)Q(-x),$$

and define

$$h_0(s) = P(z)e^{-Az^2}.$$

Then,

$$h(\gamma) = P(\eta)^2 e^{-2A\eta^2}$$

and therefore,

$$\sum_{\gamma} h(\gamma) = \sum_{\eta} P(\eta)^2 e^{-2A\eta^2}.$$

We decompose this sum as

$$\sum_{|Re(\eta)| \leq 2} + \sum_{|Re(\eta)| > 2}.$$

Let  $m$  be the order of the zero  $\rho_0$ . As  $Q$  vanishes at  $\eta$  satisfying  $|Re(\eta)| \leq 2$ , except at  $\eta_0$ , the first sum is

$$2m\eta_0^2 |Q(\eta_0)|^4 e^{-2A\eta_0^2} = -2m(\beta_0 - \frac{1}{2})^2 |Q(\eta_0)|^4 e^{2A(\beta_0 - \frac{1}{2})^2}$$

and the second sum is

$$\sum_{|Re(\eta)| > 2} P(\eta)^2 e^{-2A\eta^2} \leq e^{-2A} \sum_{|Re(\eta)| > 2} |P(\eta)^2 e^{-\eta^2}|$$

as a simple calculation reveals. Choosing  $A$  large forces the total sum to be strictly negative. That is,

$$\sum_{\gamma} h(\gamma) < 0,$$

contrary to hypothesis. This completes the proof of the corollary.

As was done by Weil, the explicit formula can be stated in the more general context of  $L$  functions satisfying suitable functional equations and growth conditions. This leads to a variety of applications. A central problem of number theory is the analytic continuation of non-abelian Artin  $L$  series. The positivity of a certain family of sums analogous to the one in the proposition above yields Artin's conjecture on the holomorphy of  $L$  series and the analogue of the Riemann hypothesis for them. The fundamental work of Langlands has shown that certain  $L$ -series attached to two dimensional representations are entire. These  $L$  series are not monomial representations and thus provide new cases of Artin's conjecture.

There is also a fundamental inequality discovered by V. Kumar Murty that relates the poles and zeroes of Artin  $L$  series to the zeroes of the (Dedekind) zeta function of a number field. Let  $L/K$  be a finite Galois extension with Galois group  $G$ . Let  $\chi$  be an irreducible character of  $G$  and denote by  $n_{\chi}(\rho)$  the order of zero or pole at  $s = \rho$  of  $L(s, \chi)$ . Let  $r(\rho)$  denote the order of zero at  $s = \rho$  of  $\zeta_L(s)$ , the Dedekind zeta function of  $L$ . Then:

**THEOREM 3.** (V. Kumar Murty) For  $\rho \neq 1$ ,

$$\sum_{\chi} n_{\chi}(\rho)^2 \leq r(\rho)^2.$$

The summation above is over irreducible characters of  $G$ . A striking consequence of this inequality is that if the Riemann hypothesis is true for the Dedekind zeta function, then the (non-abelian) Artin  $L$  series are analytic in the half plane  $\Re(s) > 1/2$ .

There is a plethora of zeta functions and  $L$ -series arising from algebraic geometry and automorphic representations of reductive algebraic groups. Perhaps, in some imperceptible way, their commingling effect will shed some light on the central problems of number theory.